

many other ways make the design and development of embedded systems a cakewalk.

Support for new standards

Here is a typical example of how embedded design and development tools have to advance in sync with other technologies, in order to enable developers to make use of the latest technologies in their embedded systems.

Tektronix recently unveiled new interspersers for its TLA7000 series logic analysers. This brings to engineers bus capture and analysis abilities for the DDR3-1867 standard—the latest generation of double-data-rate (DDR) memory technology. The company also claims to have announced the industry's first electrical test verification for LP-DDR2 – a new energy-efficient version of the DDR standard. This makes Tektronix's TLA2000 series, along with tools released by partner Nexus Technologies, a complete debug and validation suite for DDR3.

The DDR suite comes with a set of software components that enable engineers to quickly and easily set up the logic analyser to reliably capture DDR3 bus data. The software also provides a decoded view of the DDR bus activity and information about any violation in the DDR bus protocol and how analogue anomalies affect DDR memory.

The latest in software too

Just as the latest standards get reflected in hardware tools, so do all the recent developments in the world of software. Normally, most hardware tools are not standalone units, but a package or kit of the hardware along with support

software. Development boards, for instance, might either be platform-independent, enabling the use of any real-time operating system, libraries and stacks, or come with a specific software suite. In either case, all recent advancements in software are reflected in development and evaluation kits.

Take, for example, Microsoft's Silverlight technology—a development that was spoken about a lot last year. The period also saw several board support packages (BSPs) incorporating the feature.

In September last year, Bsquare announced a Windows Embedded CE 6.0 R3 board support package for the OMAP35x evaluation module from Texas Instruments (TI). The BSP provided support for the Silverlight technology, as an out-of-browser, native code implementation that could be used by developers to create rich device user interfaces including impressive high-frame-rate graphics and embedded video. CE 6.0 R3 also includes support for Adobe Flash Lite, touch and gesture support for natural input, a new multi-interface connection manager, etc.

Following this, other board support packages also began supporting the CE 6.0 R3. There came, for example, a Windows CE 6.0 R3 implementation for the SwiftModule-OM system-on-module (SOM) based on TI's OMAP3530 family devices, equipped with ARM's top-of-the-range Cortex-A8 core, a PowerVR OpenGL ES 2.0 graphics engine and C65x+ DSP. The new BSP linked Silverlight to the OMAP3530's graphics engine, enabling several desirable features including smooth pan and zoom.

SOMs such as SwiftModule enable very short time-to-market. The SOM consists of high-speed, high-density electronics, which is abstracted using a simple BSP, making it all ready to be designed into a Windows CE-based system to achieve a cost-effective implementation and rapid time-to-market. Using such modules, executing a complex project becomes as simple as creating a baseboard from the provided intellectual property (IP). The time-to-market averages 8-10 weeks only!

Enabling what the consumers want

SOMs and boards also evolve to support all the new features that are hot amongst consumers—geographical positioning systems (GPS), for example.

Last year, SwiftModule launched SwiftGPS—a production-ready GPS design-in for the SwiftModule baseboard. It includes a daughterboard that plugs into the extension header of the SwiftKit development system, making it very easy for the developer to incorporate GPS functionality into any device.

SwiftGPS uses a Global Top FGPM-MOPA6 66-channel GPS engine board with GPS antenna and provides high sensitivity. Position accuracy is less than 3 metres, and power consumption is less than 37 mA while tracking.

Similarly, there are daughterboards from various companies that facilitate features like network-attached storage and high-definition video support.

Emulators rise to the occasion

Emulators are basically pieces of hardware that imitate the behaviour

